Evaluation runs: ironlock (ironlock)

generated 2026-11-08 from brain-data 34d8e8fae 2026-11-08 19:23:20 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	159	4	conforms	0.0	4	1 h, 250 GB	4	91	brain03, 2026-11-02
3	412	4	conforms	0.0	4	1 h, 250 GB	4	149	brain03, 2026-11-02
4	845	4	conforms	0.1	4	1 h, 250 GB	4	169	brain03, 2026-11-02
5	1506	4	conforms	0.1	4	1 h, 250 GB	4	189	brain03, 2026-11-02
6	2443	4	conforms	0.2	4	1 h, 250 GB	4	209	brain03, 2026-11-02
7	3704	4	conforms	0.3	4	1 h, 250 GB	4	229	brain03, 2026-11-02
8	5337	4	conforms	0.5	4	1 h, 250 GB	4	249	brain03, 2026-11-02
9	7390	4	conforms	0.8	4	1 h, 250 GB	4	269	brain03, 2026-11-02
10	9911	4	conforms	1.2	4	1 h, 250 GB	4	289	brain03, 2026-11-02

stateless: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	27	1	conforms	0.0	4	1 h, 250 GB	1	22	brain03, 2026-11-02
3	64	1	conforms	0.0	4	1 h, 250 GB	1	55	brain03, 2026-11-02
4	125	1	conforms	0.0	4	1 h, 250 GB	1	108	brain03, 2026-11-02
5	216	1	conforms	0.0	4	1 h, 250 GB	1	189	brain03, 2026-11-02
6	343	1	conforms	0.1	4	1 h, 250 GB	1	304	brain03, 2026-11-02
7	512	1	conforms	0.1	2	1 h, 250 GB	1	459	brain03, 2026-11-02
8	729	1	conforms	0.1	4	1 h, 250 GB	1	660	brain03, 2026-11-02
9	1000	1	conforms	0.2	4	1 h, 250 GB	1	913	brain03, 2026-11-02
10	1331	1	conforms	0.2	4	1 h, 250 GB	1	1224	brain03, 2026-11-02

stateless: pair **grant-without-token**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	159	12	conforms	0.0	4	1 h, 250 GB	4	91	brain03, 2026-11-02
3	412	166	violation	0.8	4	1 h, 250 GB	88	30617	brain03, 2026-11-02
4	845	222	violation	1.7	4	1 h, 250 GB	106	77895	brain03, 2026-11-02
5	1506	382	violation	5.5	4	1 h, 250 GB	182	251054	brain03, 2026-11-02
6	2443	702	violation	19.9	4	1 h, 250 GB	334	775939	brain03, 2026-11-02
7	3704	1342	violation	64.2	4	1 h, 250 GB	638	2299724	brain03, 2026-11-02
8	5337	2622	violation	204.8	4	1 h, 250 GB	1246	6555641	brain03, 2026-11-02
9	7390	5182	violation	626.4	4	1 h, 250 GB	2462	18060650	brain03, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
10	9911	10302	violation	1875.1	6	1 h, 250 GB	4894	48322191	brain03, 2026-11-02

stateless: pair **grant-without-token**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	27	2	conforms	0.0	4	1 h, 250 GB	1	22	brain03, 2026-11-02
3	64	4	violation	0.0	4	1 h, 250 GB	4	247	brain03, 2026-11-02
4	125	7	violation	0.1	2	1 h, 250 GB	7	858	brain03, 2026-11-02
5	216	12	violation	0.1	4	1 h, 250 GB	12	2444	brain03, 2026-11-02
6	343	20	violation	0.3	4	1 h, 250 GB	20	6602	brain03, 2026-11-02
7	512	36	violation	0.9	4	1 h, 250 GB	36	18024	brain03, 2026-11-02
8	729	68	violation	2.8	4	1 h, 250 GB	68	48968	brain03, 2026-11-02
9	1000	132	violation	8.1	4	1 h, 250 GB	132	131148	brain03, 2026-11-02
10	1331	260	violation	18.3	4	1 h, 250 GB	260	344902	brain03, 2026-11-02

stateless: pair **double-hold**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	159	19	3	violation	0.0	4	1 h, 250 GB	10	686	brain03, 2026-11-02
3	412	29	6	violation	0.1	4	1 h, 250 GB	13	2759	brain03, 2026-11-02
4	845	49	12	violation	0.3	4	1 h, 250 GB	19	10467	brain03, 2026-11-02
5	1506	89	24	violation	1.3	2	1 h, 250 GB	31	36511	brain03, 2026-11-02
6	2443	169	48	violation	4.7	4	1 h, 250 GB	55	117671	brain03, 2026-11-02
7	3704	329	96	violation	16.0	4	1 h, 250 GB	103	356031	brain03, 2026-11-02
8	5337	649	192	violation	51.6	2	1 h, 250 GB	199	1025191	brain03, 2026-11-02
9	7390	1289	384	violation	156.9	4	1 h, 250 GB	391	2838287	brain03, 2026-11-02
10	9911	2569	768	violation	470.4	4	1 h, 250 GB	775	7612215	brain03, 2026-11-02

stateless: pair **double-hold**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	27	2	1	violation	0.0	4	1 h, 250 GB	2	49	brain03, 2026-11-02
3	64	3	2	violation	0.0	4	1 h, 250 GB	3	183	brain03, 2026-11-02
4	125	5	4	violation	0.0	4	1 h, 250 GB	5	608	brain03, 2026-11-02
5	216	9	8	violation	0.1	4	1 h, 250 GB	9	1917	brain03, 2026-11-02
6	343	17	16	violation	0.3	4	1 h, 250 GB	17	5792	brain03, 2026-11-02
7	512	33	32	violation	0.9	4	1 h, 250 GB	33	16843	brain03, 2026-11-02
8	729	65	64	violation	2.7	4	1 h, 250 GB	65	47316	brain03, 2026-11-02
9	1000	129	128	violation	8.0	4	1 h, 250 GB	129	128913	brain03, 2026-11-02
10	1331	257	256	violation	23.5	4	1 h, 250 GB	257	341960	brain03, 2026-11-02

Not run (stateless). planned but never scheduled (grid points beyond the ladder): 11 grant-without-token bag, 12 double-hold fifo, 12 grant-without-token fifo, 12 double-hold bag, 20 correct fifo, 20 correct bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	159	4	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-01
3	412	4	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-01
4	845	4	conforms	0.1	2	1 h, 250 GB	brain02, 2026-11-01
5	1506	4	conforms	0.1	2	1 h, 250 GB	brain02, 2026-11-01
6	2443	4	conforms	0.1	2	1 h, 250 GB	brain02, 2026-11-01
7	3704	4	conforms	0.2	8	1 h, 250 GB	brain02, 2026-11-01
8	5337	4	conforms	0.4	8	1 h, 250 GB	brain02, 2026-11-01

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
9	7390	4	conforms	0.5	10	1 h, 250 GB	brain02, 2026-11-01
10	9911	4	conforms	0.8	14	1 h, 250 GB	brain02, 2026-11-01

stateful: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	27	1	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-01
3	64	1	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-01
4	125	1	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-01
5	216	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-01
6	343	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-01
7	512	1	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-01
8	729	1	conforms	0.1	8	1 h, 250 GB	brain02, 2026-11-01
9	1000	1	conforms	0.1	6	1 h, 250 GB	brain02, 2026-11-01
10	1331	1	conforms	0.1	12	1 h, 250 GB	brain02, 2026-11-01

stateful: pair **grant-without-token**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	159	12	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-01
3	412	166	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-01
4	845	222	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-01
5	1506	382	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-01
6	2443	702	violation	0.2	2	1 h, 250 GB	brain02, 2026-11-01
7	3704	1342	violation	0.3	12	1 h, 250 GB	brain02, 2026-11-01
8	5337	2622	violation	0.6	14	1 h, 250 GB	brain02, 2026-11-01
9	7390	5182	violation	0.9	22	1 h, 250 GB	brain02, 2026-11-01
10	9911	10302	violation	1.7	26	1 h, 250 GB	brain02, 2026-11-01

stateful: pair **grant-without-token**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	27	2	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-01
3	64	4	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-01
4	125	7	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-01
5	216	12	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-01
6	343	20	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-01
7	512	36	violation	0.1	6	1 h, 250 GB	brain02, 2026-11-01
8	729	68	violation	0.1	6	1 h, 250 GB	brain02, 2026-11-01
9	1000	132	violation	0.1	10	1 h, 250 GB	brain02, 2026-11-01
10	1331	260	violation	0.2	12	1 h, 250 GB	brain02, 2026-11-01

stateful: pair **double-hold**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	159	19	3	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-01
3	412	29	6	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-01
4	845	49	12	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-01
5	1506	89	24	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-01
6	2443	169	48	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-01
7	3704	329	96	violation	0.2	8	1 h, 250 GB	brain02, 2026-11-01
8	5337	649	192	violation	0.4	8	1 h, 250 GB	brain02, 2026-11-01
9	7390	1289	384	violation	0.7	12	1 h, 250 GB	brain02, 2026-11-01
10	9911	2569	768	violation	1.0	12	1 h, 250 GB	brain02, 2026-11-01

stateful: pair **double-hold**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	27	2	1	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-01
3	64	3	2	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-01
4	125	5	4	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-01
5	216	9	8	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-01
6	343	17	16	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-01
7	512	33	32	violation	0.1	2	1 h, 250 GB	brain02, 2026-11-01
8	729	65	64	violation	0.1	8	1 h, 250 GB	brain02, 2026-11-01
9	1000	129	128	violation	0.1	8	1 h, 250 GB	brain02, 2026-11-01
10	1331	257	256	violation	0.2	12	1 h, 250 GB	brain02, 2026-11-01

Not run (stateful). planned but never scheduled (grid points beyond the ladder): 20 double-hold fifo, 20 double-hold bag, 20 correct fifo, 20 correct bag, 20 grant-without-token bag, 20 grant-without-token fifo.

offline

offline: pair **correct** against **correct-ring**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
5	4	4	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
6	4	4	conforms	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
7	4	4	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
8	4	4	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
9	4	4	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
10	4	4	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02

offline: pair **correct** against **correct-ring**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
5	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
6	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
7	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
8	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
9	1	1	conforms	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
10	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02

offline: pair **correct** against **ironlock.spec-ring**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	4	conforms	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain05, 2026-11-02
3	4	4	conforms	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
4	4	4	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02

offline: pair **correct** against **ironlock.spec-ring**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	1	1	conforms	0.1		10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	1	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	159	4	conforms	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
3	412	4	conforms	0.2	8	10 min, 16 GB	0.2	0.2	0.0	in-memory	brain05, 2026-11-02
4	845	4	conforms	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
5	1506	4	conforms	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
6	2443	4	conforms	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
7	3704	4	conforms	0.4	8	10 min, 16 GB	0.4	0.1	0.0	in-memory	brain05, 2026-11-02
8	5337	4	conforms	0.5	10	10 min, 16 GB	0.5	0.1	0.0	in-memory	brain05, 2026-11-02
9	7390	4	conforms	0.7	12	10 min, 16 GB	0.7	0.1	0.1	in-memory	brain05, 2026-11-02
10	9911	4	conforms	1.1	18	10 min, 16 GB	1.0	0.1	0.1	in-memory	brain05, 2026-11-02

offline: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	27	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	64	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	125	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
5	216	1	conforms	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
6	343	1	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
7	512	1	conforms	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
8	729	1	conforms	0.2	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
9	1000	1	conforms	0.2	10	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
10	1331	1	conforms	0.3	14	10 min, 16 GB	0.2	0.0	0.1	in-memory	brain05, 2026-11-02

offline: pair **grant-without-token** against **ironlock.spec-ring**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	12	conforms	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
3	4	166	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	4	222	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02

offline: pair **grant-without-token** against **ironlock.spec-ring**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	2	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	1	4	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	1	7	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02

offline: pair **grant-without-token**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	159	12	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	412	166	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
4	845	222	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
5	1506	382	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
6	2443	702	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
7	3704	1342	violation	0.4	8	10 min, 16 GB	0.4	0.2	0.0	in-memory	brain05, 2026-11-02
8	5337	2622	violation	0.5	10	10 min, 16 GB	0.5	0.3	0.0	in-memory	brain05, 2026-11-02
9	7390	5182	violation	0.7	12	10 min, 16 GB	0.7	0.5	0.1	in-memory	brain05, 2026-11-02
10	9911	10302	violation	1.2	18	10 min, 16 GB	1.0	1.1	0.1	in-memory	brain05, 2026-11-02

offline: pair **grant-without-token**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	27	2	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	64	4	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	125	7	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
5	216	12	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
6	343	20	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
7	512	36	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
8	729	68	violation	0.2	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
9	1000	132	violation	0.2	10	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
10	1331	260	violation	0.3	12	10 min, 16 GB	0.2	0.2	0.1	in-memory	brain05, 2026-11-02

offline: pair **double-hold** against **correct-ring**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
5	4	89	6	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
6	4	169	6	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
7	4	329	6	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
8	4	649	6	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
9	4	1289	6	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
10	4	2569	6	violation	0.5	8	10 min, 16 GB	0.1	0.5	0.0	in-memory	brain05, 2026-11-02

offline: pair **double-hold** against **correct-ring**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
5	1	9	1	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
6	1	17	1	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
7	1	33	1	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
8	1	65	1	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
9	1	129	1	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
10	1	257	1	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02

offline: pair **double-hold** against **ironlock.spec-ring**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	19	6	violation	0.1	8	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
3	4	29	6	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	4	49	6	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02

offline: pair **double-hold** against **ironlock.spec-ring**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	2	1	violation	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
3	1	3	1	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	1	5	1	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02

offline: pair **double-hold**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	159	19	3	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3	412	29	3	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
4	845	49	3	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
5	1506	89	3	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
6	2443	169	3	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
7	3704	329	3	violation	0.4	8	10 min, 16 GB	0.4	0.1	0.0	in-memory	brain05, 2026-11-02
8	5337	649	3	violation	0.5	10	10 min, 16 GB	0.5	0.1	0.0	in-memory	brain05, 2026-11-02
9	7390	1289	3	violation	0.7	12	10 min, 16 GB	0.7	0.2	0.1	in-memory	brain05, 2026-11-02
10	9911	2569	3	violation	1.1	18	10 min, 16 GB	1.0	0.5	0.1	in-memory	brain05, 2026-11-02

offline: pair **double-hold**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	27	2	1	violation	0.1	8	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
3	64	3	1	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	125	5	1	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
5	216	9	1	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
6	343	17	1	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
7	512	33	1	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
8	729	65	1	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
9	1000	129	1	violation	0.2	10	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
10	1331	257	1	violation	0.3	14	10 min, 16 GB	0.2	0.2	0.1	in-memory	brain05, 2026-11-02

offline: pair **grant-without-token** against **correct-ring**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
5	4	382	36	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
6	4	702	36	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
7	4	1342	36	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
8	4	2622	36	violation	0.3	8	10 min, 16 GB	0.1	0.3	0.0	in-memory	brain05, 2026-11-02
9	4	5182	36	violation	0.5	8	10 min, 16 GB	0.1	0.5	0.0	in-memory	brain05, 2026-11-02
10	4	10302	36	violation	1.1	8	10 min, 16 GB	0.1	1.1	0.0	in-memory	brain05, 2026-11-02

offline: pair **grant-without-token** against **correct-ring**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
5	1	12	4	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
6	1	20	4	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
7	1	36	4	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
8	1	68	4	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
9	1	132	4	violation	0.2	8	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
10	1	260	4	violation	0.2	8	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02

Not run (offline). planned but never scheduled (grid points beyond the ladder): 20 double-hold fifo, 20 double-hold bag, 20 correct fifo, 20 correct bag, 20 grant-without-token bag, 20 grant-without-token fifo.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	159	4	conforms	2.9	173	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-02
3	412	4	conforms	3.8	162	10 min, 16 GB	0.2	3.8	0.0	in-memory	brain01, 2026-11-02
4	845	4	conforms	3.8	160	10 min, 16 GB	0.2	3.8	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	27	1	conforms	3.6	167	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain01, 2026-11-02
3	64	1	conforms	3.8	161	10 min, 16 GB	0.1	3.8	0.0	in-memory	brain01, 2026-11-02
4	125	1	conforms	3.6	160	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **grant-without-token**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	159	12	conforms	3.1	158	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
3	412	166	violation	3.8	173	10 min, 16 GB	0.2	3.8	0.0	in-memory	brain01, 2026-11-02
4	845	222	violation	4.0	179	10 min, 16 GB	0.2	4.0	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **grant-without-token**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	27	2	conforms	2.9	160	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-02
3	64	4	violation	3.1	176	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
4	125	7	violation	3.1	170	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **double-hold**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	159	19	3	violation	3.1	171	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
3	412	29	3	violation	3.1	156	10 min, 16 GB	0.2	3.1	0.0	in-memory	brain01, 2026-11-02
4	845	49	3	violation	3.4	158	10 min, 16 GB	0.2	3.4	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **double-hold**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	27	2	1	violation	2.9	174	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-02
3	64	3	1	violation	3.1	176	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
4	125	5	1	violation	4.0	167	10 min, 16 GB	0.1	4.0	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Java implementation against the Rust specification). planned but never scheduled (grid points beyond the ladder): 5 fifo correct, 5 bag double-hold, 5 bag correct, 5 fifo double-hold, 5 fifo grant-without-token, 5 bag grant-without-token.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	159	4	conforms	3.3	172	10 min, 16 GB	3.3	0.0	0.0	in-memory	brain01, 2026-11-02
3	412	4	conforms	3.5	166	10 min, 16 GB	3.5	0.2	0.0	in-memory	brain01, 2026-11-02
4	845	4	conforms	4.0	158	10 min, 16 GB	4.0	0.1	0.1	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	27	1	conforms	3.1	167	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
3	64	1	conforms	3.1	166	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
4	125	1	conforms	4.0	160	10 min, 16 GB	4.0	0.1	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **grant-without-token**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	159	12	conforms	3.3	172	10 min, 16 GB	3.3	0.1	0.0	in-memory	brain01, 2026-11-02
3	412	166	violation	3.5	166	10 min, 16 GB	3.5	0.1	0.0	in-memory	brain01, 2026-11-02
4	845	222	violation	4.0	158	10 min, 16 GB	4.0	0.1	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **grant-without-token**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	27	2	conforms	3.1	167	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
3	64	4	violation	3.1	166	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
4	125	7	violation	4.0	160	10 min, 16 GB	4.0	0.1	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **double-hold**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	159	19	3	violation	3.3	172	10 min, 16 GB	3.3	0.1	0.0	in-memory	brain01, 2026-11-02
3	412	29	3	violation	3.5	166	10 min, 16 GB	3.5	0.1	0.0	in-memory	brain01, 2026-11-02
4	845	49	3	violation	4.0	158	10 min, 16 GB	4.0	0.1	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair **double-hold**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	27	2	1	violation	3.1	167	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
3	64	3	1	violation	3.1	166	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
4	125	5	1	violation	4.0	160	10 min, 16 GB	4.0	0.1	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 5 fifo correct, 5 bag double-hold, 5 bag correct, 5 fifo double-hold, 5 fifo grant-without-token, 5 bag grant-without-token.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	159	4	conforms	3.3	168	1 h, 250 GB		brain01, 2026-11-02
3	412	4	conforms	3.5	162	1 h, 250 GB		brain01, 2026-11-02
4	845	4	conforms	4.0	154	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	27	1	conforms	3.1	163	1 h, 250 GB		brain01, 2026-11-02
3	64	1	conforms	3.1	162	1 h, 250 GB		brain01, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
4	125	1	conforms	4.0	156	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **grant-without-token**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	159	12	conforms	3.4	168	1 h, 250 GB		brain01, 2026-11-02
3	412	166	violation	3.6	162	1 h, 250 GB	2	brain01, 2026-11-02
4	845	222	violation	4.0	154	1 h, 250 GB	2	brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **grant-without-token**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	27	2	conforms	3.1	163	1 h, 250 GB		brain01, 2026-11-02
3	64	4	violation	3.1	162	1 h, 250 GB	2	brain01, 2026-11-02
4	125	7	violation	4.0	156	1 h, 250 GB	2	brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **double-hold**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	159	19	3	violation	3.4	168	1 h, 250 GB	1	brain01, 2026-11-02
3	412	29	6	violation	3.5	162	1 h, 250 GB	2	brain01, 2026-11-02
4	845	49	12	violation	4.0	154	1 h, 250 GB	3	brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **double-hold**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	27	2	1	violation	3.1	163	1 h, 250 GB	2	brain01, 2026-11-02
3	64	3	2	violation	3.1	162	1 h, 250 GB	2	brain01, 2026-11-02
4	125	5	4	violation	4.0	156	1 h, 250 GB	2	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	4	conforms	2.9	173	10 min, 16 GB	2.9	0.0	0.0	in-memory	brain01, 2026-11-02
3	4	4	conforms	3.8	162	10 min, 16 GB	3.8	0.2	0.0	in-memory	brain01, 2026-11-02
4	4	4	conforms	3.8	160	10 min, 16 GB	3.8	0.1	0.1	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	1	conforms	3.6	167	10 min, 16 GB	3.6	0.1	0.1	in-memory	brain01, 2026-11-02
3	1	1	conforms	3.8	161	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
4	1	1	conforms	3.6	160	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair double-hold, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	19	19	conforms	3.1	171	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
3	29	29	conforms	3.1	157	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
4	49	49	conforms	3.4	157	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair double-hold, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	2	2	conforms	2.9	173	10 min, 16 GB	2.9	0.0	0.0	in-memory	brain01, 2026-11-02
3	3	3	conforms	3.1	176	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
4	5	5	conforms	4.0	167	10 min, 16 GB	4.0	0.1	0.1	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair grant-without-token, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	12	conforms	3.1	159	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
3	166	166	conforms	3.8	173	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02
4	222	222	conforms	4.2	179	10 min, 16 GB	4.0	0.1	0.2	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair grant-without-token, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	2	2	conforms	3.2	160	10 min, 16 GB	2.9	0.1	0.2	in-memory	brain01, 2026-11-02
3	4	4	conforms	3.1	176	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
4	7	7	conforms	3.1	170	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair double-hold, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	19	6	violation	2.9	173	10 min, 16 GB	2.9	0.1	0.0	in-memory	brain01, 2026-11-02
3	4	29	6	violation	3.8	162	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02
4	4	49	6	violation	3.8	160	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair double-hold, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	2	1	violation	3.6	167	10 min, 16 GB	3.6	0.0	0.0	in-memory	brain01, 2026-11-02
3	1	3	1	violation	3.8	161	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02
4	1	5	1	violation	3.6	160	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **grant-without-token**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	12		conforms*	2.9	173	10 min, 16 GB	2.9	0.1	0.0	in-memory	brain01, 2026-11-02
3	4	166	46	violation	3.8	162	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02
4	4	222	36	violation	3.8	160	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **grant-without-token**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	2		conforms*	3.6	167	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02
3	1	4	2	violation	3.8	161	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02
4	1	7	3	violation	3.6	160	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02

Not run (twins offline, Rust implementation against its Java twin). planned but never scheduled (grid points beyond the ladder): 5 fifo correct, 5 bag double-hold, 5 bag correct, 5 fifo double-hold, 5 fifo grant-without-token, 5 bag grant-without-token.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	4	conforms	2.9	173	10 min, 16 GB	0.0	2.9	0.0	in-memory	brain01, 2026-11-02
3	4	4	conforms	3.8	162	10 min, 16 GB	0.2	3.8	0.0	in-memory	brain01, 2026-11-02
4	4	4	conforms	3.8	160	10 min, 16 GB	0.1	3.8	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	1	conforms	3.6	167	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain01, 2026-11-02
3	1	1	conforms	3.8	161	10 min, 16 GB	0.1	3.8	0.0	in-memory	brain01, 2026-11-02
4	1	1	conforms	3.6	160	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **double-hold**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	19	19	conforms	3.1	171	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
3	29	29	conforms	3.1	157	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
4	49	49	conforms	3.4	157	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **double-hold**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	2	2	conforms	2.9	173	10 min, 16 GB	0.0	2.9	0.0	in-memory	brain01, 2026-11-02
3	3	3	conforms	3.1	176	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
4	5	5	conforms	4.0	167	10 min, 16 GB	0.1	4.0	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair grant-without-token, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	12	conforms	3.1	159	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
3	166	166	conforms	3.8	173	10 min, 16 GB	0.1	3.8	0.0	in-memory	brain01, 2026-11-02
4	222	222	conforms	4.0	179	10 min, 16 GB	0.1	4.0	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair grant-without-token, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	2	2	conforms	2.9	160	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-02
3	4	4	conforms	3.1	176	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
4	7	7	conforms	3.1	170	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair double-hold, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	19	6	violation	3.1	171	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
3	4	29	6	violation	3.1	156	10 min, 16 GB	0.2	3.1	0.0	in-memory	brain01, 2026-11-02
4	4	49	6	violation	3.4	158	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair double-hold, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	2	1	violation	2.9	174	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-02
3	1	3	1	violation	3.1	176	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
4	1	5	1	violation	4.0	167	10 min, 16 GB	0.1	4.0	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair grant-without-token, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	12		conforms*	3.1	158	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
3	4	166	46	violation	3.8	173	10 min, 16 GB	0.2	3.8	0.0	in-memory	brain01, 2026-11-02
4	4	222	36	violation	4.0	179	10 min, 16 GB	0.1	4.0	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair grant-without-token, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	2		conforms*	2.9	160	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-02
3	1	4	2	violation	3.1	176	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
4	1	7	3	violation	3.1	170	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02

Not run (twins offline, Java implementation against its Rust twin). planned but never scheduled (grid points beyond the ladder): 5 fifo correct, 5 bag double-hold, 5 bag correct, 5 fifo double-hold, 5 fifo grant-without-token, 5 bag grant-without-token.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	4	conforms	3.1	174	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain02, 2026-11-03
3	4	4	conforms	3.5	162	10 min, 16 GB	3.3	0.1	0.2	in-memory	brain02, 2026-11-03
4	4	4	conforms	3.3	160	10 min, 16 GB	3.3	0.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	1	conforms	3.5	156	10 min, 16 GB	2.9	0.0	0.6	in-memory	brain02, 2026-11-03
3	1	1	conforms	3.1	156	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain02, 2026-11-03
4	1	1	conforms	3.0	163	10 min, 16 GB	2.9	0.2	0.1	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **double-hold**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	19	19	conforms	3.9	170	10 min, 16 GB	3.3	0.1	0.5	in-memory	brain02, 2026-11-03
3	29	29	conforms	3.3	158	10 min, 16 GB	3.3	0.0	0.0	in-memory	brain02, 2026-11-03
4	49	49	conforms	3.6	156	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **double-hold**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	2	2	conforms	3.1	164	10 min, 16 GB	3.1	0.2	0.0	in-memory	brain02, 2026-11-03
3	3	3	conforms	2.9	156	10 min, 16 GB	2.9	0.1	0.0	in-memory	brain02, 2026-11-03
4	5	5	conforms	3.3	162	10 min, 16 GB	3.3	0.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **grant-without-token**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	12	conforms	3.1	158	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain02, 2026-11-03
3	166	166	conforms	3.8	178	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain02, 2026-11-03
4	222	222	conforms	4.1	164	10 min, 16 GB	4.1	0.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **grant-without-token**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	2	2	conforms	3.1	165	10 min, 16 GB	2.9	0.1	0.2	in-memory	brain02, 2026-11-03
3	4	4	conforms	3.3	157	10 min, 16 GB	3.3	0.1	0.0	in-memory	brain02, 2026-11-03
4	7	7	conforms	3.7	170	10 min, 16 GB	3.3	0.1	0.4	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **double-hold**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	19	6	violation	3.1	174	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain02, 2026-11-03
3	4	29	6	violation	3.3	162	10 min, 16 GB	3.3	0.0	0.0	in-memory	brain02, 2026-11-03
4	4	49	6	violation	3.3	160	10 min, 16 GB	3.3	0.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **double-hold**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	2	1	violation	2.9	156	10 min, 16 GB	2.9	0.2	0.0	in-memory	brain02, 2026-11-03
3	1	3	1	violation	3.1	156	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain02, 2026-11-03
4	1	5	1	violation	2.9	163	10 min, 16 GB	2.9	0.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **grant-without-token**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	12		conforms*	3.1	174	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain02, 2026-11-03
3	4	166	46	violation	3.3	162	10 min, 16 GB	3.3	0.1	0.0	in-memory	brain02, 2026-11-03
4	4	222	36	violation	3.3	160	10 min, 16 GB	3.3	0.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **grant-without-token**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	2		conforms*	2.9	156	10 min, 16 GB	2.9	0.1	0.0	in-memory	brain02, 2026-11-03
3	1	4	2	violation	3.1	157	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain02, 2026-11-03
4	1	7	3	violation	2.9	163	10 min, 16 GB	2.9	0.1	0.0	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). planned but never scheduled (grid points beyond the ladder): 5 fifo correct, 5 bag double-hold, 5 bag correct, 5 fifo double-hold, 5 fifo grant-without-token, 5 bag grant-without-token.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	4	conforms	3.1	174	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain02, 2026-11-03
3	4	4	conforms	3.5	162	10 min, 16 GB	0.1	3.3	0.2	in-memory	brain02, 2026-11-03
4	4	4	conforms	3.3	160	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	1	conforms	3.5	156	10 min, 16 GB	0.0	2.9	0.6	in-memory	brain02, 2026-11-03
3	1	1	conforms	3.1	156	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain02, 2026-11-03
4	1	1	conforms	3.0	163	10 min, 16 GB	0.2	2.9	0.1	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair double-hold, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	19	19	conforms	3.8	170	10 min, 16 GB	0.1	3.3	0.5	in-memory	brain02, 2026-11-03
3	29	29	conforms	3.3	158	10 min, 16 GB	0.0	3.3	0.0	in-memory	brain02, 2026-11-03
4	49	49	conforms	3.6	156	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair double-hold, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	2	2	conforms	3.1	164	10 min, 16 GB	0.2	3.1	0.0	in-memory	brain02, 2026-11-03
3	3	3	conforms	2.9	156	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain02, 2026-11-03
4	5	5	conforms	3.3	162	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair grant-without-token, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	12	conforms	3.1	158	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain02, 2026-11-03
3	166	166	conforms	3.8	178	10 min, 16 GB	0.1	3.8	0.0	in-memory	brain02, 2026-11-03
4	222	222	conforms	4.1	164	10 min, 16 GB	0.1	4.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair grant-without-token, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	2	2	conforms	3.0	165	10 min, 16 GB	0.1	2.9	0.2	in-memory	brain02, 2026-11-03
3	4	4	conforms	3.3	157	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain02, 2026-11-03
4	7	7	conforms	3.7	170	10 min, 16 GB	0.1	3.3	0.4	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair double-hold, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	19	6	violation	3.3	170	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain02, 2026-11-03
3	4	29	6	violation	3.3	158	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain02, 2026-11-03
4	4	49	6	violation	3.6	156	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair double-hold, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	2	1	violation	3.1	164	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain02, 2026-11-03
3	1	3	1	violation	2.9	156	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain02, 2026-11-03
4	1	5	1	violation	3.3	162	10 min, 16 GB	0.2	3.3	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair grant-without-token, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	12		conforms*	3.1	158	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain02, 2026-11-03
3	4	166	46	violation	3.8	178	10 min, 16 GB	0.1	3.8	0.0	in-memory	brain02, 2026-11-03
4	4	222	36	violation	4.1	164	10 min, 16 GB	0.1	4.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **grant-without-token**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	1	2		conforms*	2.9	165	10 min, 16 GB	0.0	2.9	0.0	in-memory	brain02, 2026-11-03
3	1	4	2	violation	3.3	157	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain02, 2026-11-03
4	1	7	3	violation	3.3	170	10 min, 16 GB	0.2	3.3	0.0	in-memory	brain02, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). planned but never scheduled (grid points beyond the ladder): 5 fifo correct, 5 bag double-hold, 5 bag correct, 5 fifo double-hold, 5 fifo grant-without-token, 5 bag grant-without-token.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	4	4	conforms	3.9	162	1 h, 250 GB		brain02, 2026-11-03
3	4	4	conforms	3.4	154	1 h, 250 GB		brain02, 2026-11-03
4	4	4	conforms	3.5	156	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	1	1	conforms	3.4	153	1 h, 250 GB		brain02, 2026-11-03
3	1	1	conforms	3.5	156	1 h, 250 GB		brain02, 2026-11-03
4	1	1	conforms	4.0	170	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **double-hold**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	19	19	conforms	4.0	154	1 h, 250 GB		brain02, 2026-11-03
3	29	29	conforms	3.9	167	1 h, 250 GB		brain02, 2026-11-03
4	49	49	conforms	4.2	161	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **double-hold**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	2	2	conforms	3.9	160	1 h, 250 GB		brain02, 2026-11-03
3	3	3	conforms	3.3	168	1 h, 250 GB		brain02, 2026-11-03
4	5	5	conforms	3.4	168	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair grant-without-token, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	12	12	conforms	4.7	158	1 h, 250 GB		brain02, 2026-11-03
3	166	166	conforms	4.7	170	1 h, 250 GB		brain02, 2026-11-03
4	222	222	conforms	5.0	168	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair grant-without-token, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	2	2	conforms	3.4	158	1 h, 250 GB		brain02, 2026-11-03
3	4	4	conforms	3.5	173	1 h, 250 GB		brain02, 2026-11-03
4	7	7	conforms	3.7	159	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair double-hold, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	4	19	9	violation	3.8	156	1 h, 250 GB	72	brain02, 2026-11-03
3	4	29	15	violation	3.4	158	1 h, 250 GB	74	brain02, 2026-11-03
4	4	49	27	violation	4.0	159	1 h, 250 GB	84	brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair double-hold, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	1	2	1	violation	3.4	154	1 h, 250 GB	73	brain02, 2026-11-03
3	1	3	2	violation	3.8	165	1 h, 250 GB	81	brain02, 2026-11-03
4	1	5	4	violation	3.8	166	1 h, 250 GB	87	brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair grant-without-token, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	4	12		conforms*	3.7	165	1 h, 250 GB		brain02, 2026-11-03
3	4	166	139	violation	4.1	166	1 h, 250 GB	85	brain02, 2026-11-03
4	4	222	187	violation	4.6	160	1 h, 250 GB	79	brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair grant-without-token, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	1	2		conforms*	3.7	179	1 h, 250 GB		brain02, 2026-11-03
3	1	4	3	violation	3.8	151	1 h, 250 GB	89	brain02, 2026-11-03
4	1	7	6	violation	3.9	160	1 h, 250 GB	90	brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	4	4	conforms	3.3	170	1 h, 250 GB		brain02, 2026-11-03
3	4	4	conforms	3.3	158	1 h, 250 GB		brain02, 2026-11-03
4	4	4	conforms	3.3	156	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	1	1	conforms	2.9	152	1 h, 250 GB		brain02, 2026-11-03
3	1	1	conforms	3.1	152	1 h, 250 GB		brain02, 2026-11-03
4	1	1	conforms	2.9	158	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **double-hold**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	19	19	conforms	3.3	166	1 h, 250 GB		brain02, 2026-11-03
3	29	29	conforms	3.4	154	1 h, 250 GB		brain02, 2026-11-03
4	49	49	conforms	3.6	152	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **double-hold**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	2	2	conforms	3.1	160	1 h, 250 GB		brain02, 2026-11-03
3	3	3	conforms	2.9	152	1 h, 250 GB		brain02, 2026-11-03
4	5	5	conforms	3.3	158	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **grant-without-token**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	12	12	conforms	3.1	154	1 h, 250 GB		brain02, 2026-11-03
3	166	166	conforms	3.8	174	1 h, 250 GB		brain02, 2026-11-03
4	222	222	conforms	4.2	160	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **grant-without-token**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	2	2	conforms	2.9	161	1 h, 250 GB		brain02, 2026-11-03
3	4	4	conforms	3.3	153	1 h, 250 GB		brain02, 2026-11-03
4	7	7	conforms	3.3	166	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair double-hold, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	4	19	9	violation	3.1	170	1 h, 250 GB	1	brain02, 2026-11-03
3	4	29	15	violation	3.3	158	1 h, 250 GB	1	brain02, 2026-11-03
4	4	49	27	violation	3.3	156	1 h, 250 GB	1	brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair double-hold, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	1	2	1	violation	2.9	152	1 h, 250 GB	1	brain02, 2026-11-03
3	1	3	2	violation	3.1	152	1 h, 250 GB	1	brain02, 2026-11-03
4	1	5	4	violation	2.9	158	1 h, 250 GB	1	brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair grant-without-token, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	4	12		conforms*	3.1	170	1 h, 250 GB		brain02, 2026-11-03
3	4	166	139	violation	3.3	158	1 h, 250 GB	2	brain02, 2026-11-03
4	4	222	187	violation	3.3	156	1 h, 250 GB	2	brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair grant-without-token, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	1	2		conforms*	2.9	152	1 h, 250 GB		brain02, 2026-11-03
3	1	4	3	violation	3.1	152	1 h, 250 GB	1	brain02, 2026-11-03
4	1	7	6	violation	2.9	158	1 h, 250 GB	1	brain02, 2026-11-03